

FRANZ KESTER P. CALUMPIT

BS Computer Science Student | Aspiring Software Developer

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SUMMARY

Upcoming 4th year BS Computer Science student at the Polytechnic University of the Philippines with a consistent President's Lister academic record and hands-on experience building full-stack and AI-integrated applications. Active in several developer and cloud communities. Seeking a software development internship to apply and grow technical skills on real-world systems.

TECHNICAL SKILLS

Languages: Java, C#, Python, JavaScript, PHP, SQL, HTML/CSS

Frameworks & Tools: Spring Boot (JPA, Spring Security, JWT), ASP.NET Core MVC, Entity Framework Core, .NET MAUI / Blazor, React, Vite, Tailwind CSS, Unity 6

AI / ML: PyTorch, Reinforcement Learning (Deep Q-Networks), Google Gemini API integration, ONNX model inference, A* pathfinding

Databases: PostgreSQL, SQL Server, MySQL, H2; relational design (ERD), REST APIs

Cloud & Tools: Supabase, Vercel, Git & GitHub, Maven, Apache PDFBox, Agile, documentation

PROJECTS

RouteSync (BGC Bus)

2026

Mobile camera-based passenger counting and bus tracking system with dispatch management and reporting, serving a web dashboard and a mobile client.

- Co-developed backend services in C# with ASP.NET Core MVC and Entity Framework Core (SQL Server), plus a .NET MAUI mobile client and Supabase integration.
- Designed the relational database schema (ERD, ~9 tables covering trips, telemetry, and vehicle checklists) and built PDF reporting with QuestPDF.
- Collaborated in a 4-member team on system design, risk assessment, and technical documentation.

2PAC - 2-Player Cooperative Pac-Man

2026

A two-player co-op reimagining of Pac-Man for mobile, where tethered players survive reinforcement-learning ghosts across three maps using charge-based abilities and paired power-ups.

- Built ghost AI using a Deep Q-Network (Dueling + Double DQN) trained in Python with PyTorch, exported to ONNX and run natively in Unity via the AI Inference Engine.
- Implemented Weighted A* pathfinding with dynamic node costs so ghosts route around danger zones automatically.
- Developed core gameplay in C# (Unity 6, URP 2D): tether states, charge-based shockwave, paired power-ups, and 3 maps.

LUMA — Custom OOP Programming Language & Web Analyzer

2026

A beginner-friendly object-oriented language with a web-based analyzer that tokenizes and validates LUMA source code, blending Python's readability with Java's structure.

- Built the lexical and syntax analysis backend in Java 21 with Spring Boot 3.5.8 (Maven), exposing REST endpoints that tokenize source code and validate it against a grammar registry of production rules.
- Designed LUMA's language specification — character set, identifiers, operators, keywords, and OOP semantics (encapsulation via shield, inheritance via shared, and multi-view polymorphism) with finite-automata and regular-expression definitions.
- Developed a React 19 front end for live code analysis and authored a full syntax test suite, contributing to language design and technical documentation in a 7-member team.

MetaSmish — CNN–BiGRU Smishing (SMS Phishing) Detector

2026

A hybrid deep-learning + ensemble system that classifies SMS messages as smishing or legitimate, combining a from-scratch CNN–BiGRU text model with a URL-feature Random Forest, served through a web classifier.

- Implemented a hybrid CNN–BiGRU neural network from scratch in NumPy — full forward and backpropagation for Embedding, Conv1D, MaxPooling, Dropout, bidirectional GRU cells, and Dense layers — rather than relying on a high-level framework.

- Built a weighted soft-voting ensemble combining the CNN–BiGRU text classifier with a scikit-learn Random Forest trained on 11 engineered URL features (length, subdomain count, URL shorteners, IP-address detection, suspicious keywords, HTTPS).
- Engineered the NLP preprocessing and URL pipeline in Python with NLTK (tokenization, stop-word removal, padding) plus regex-based URL extraction and a trusted-domain allowlist tuned for Philippine banks, e-wallets, and government sites.
- Developed a Flask web app (HTML/CSS) exposing an interactive classifier that returns the final label, per-model probabilities, and the soft-voting breakdown.

Resume Roaster — AI Resume Analyzer

2025

Full-stack tool that gives honest, AI-generated feedback on technical resumes.

- Integrated the Google Gemini API into a Spring Boot (Java 17) backend to analyze uploaded resumes and return structured, actionable feedback.
- Built a React + Tailwind CSS front end with history tracking and PDF export (Apache PDFBox); persisted analyses in PostgreSQL.

EDUCATION

Bachelor of Science in Computer Science

Sept 2023 – Present

Polytechnic University of the Philippines, Manila

- President's Lister all five semesters to date (GWA range 1.14 – 1.46).

Junior & Senior High School

2017 – 2023

University of Perpetual Help Manila

- Graduated With High Honors (Grade 12 GWA: 96; Grade 11 GWA: 95).

LEADERSHIP & ORGANIZATIONS

Google Developer Groups – PUP

2023 – Present

Logistics Associate / Web Development Project Analyst

- Coordinated logistics and event materials for developer-focused events; contributed to web development team projects.

PUP AWS Cloud Club & AWSUG e:Novators PH

2023 – Present

Program Associate

- Supported event planning, technical operations, project management, and media coverage for technical and educational programs.

AWARDS, HONORS & EXTRACURRICULAR

- PyTorch Philippines Community Launch & AI Meetup Technicals June 2026
- Third Year 1st Sem President's Lister – GWA: 1.46 Jan 2026
- Second Year 2nd Sem President's Lister – GWA: 1.32 July 2025
- AWSCCPUP Hackathon – Technical Volunteer May 2025
- AWS Student Community Day 2025 Mega Manila – Core Member Apr – Oct 2025
- Next Hire Philippine Tech Career Fest – Ambassador Jan – Mar 2025
- AWSCC e:Novators – Member Jan 2025 – Present
- Second Year 1st Sem President's Lister – GWA: 1.14 Jan 2025
- TEDx 2024 "Anchored in Tech: Lifestyle, Leadership, and Legacy" – Attendee Nov 2024
- Google Developer Groups – Web Development Team Project Analyst Nov 2024 – Present
- AWS Cloud Club Skill Builder Program – Web Development Department Oct 2024 – Present
- First Year 2nd Sem President's Lister – GWA: 1.21 July 2024
- AWS CodeQuest: 30 Days of Front-end Development – Certificate of Completion June 2024
- AWS CC Student Community Day PH – Certificate of Recognition Apr 2024
- AWS CC Student Re:Invent Re:Cap – Certificate of Participation Mar 2024
- Collabothon: Idea Jam for the Dept. of Budget and Management – Certificate of Appreciation Feb 2024
- AWS Services Exploration – Certificate of Participation Feb 2024
- First Year 1st Sem President's Lister – GWA: 1.33 Jan 2024
- Git It Done: Git and GitHub Workshop – Certificate of Completion Jan 2024
- Grade 12 With High Honors – GWA: 96 Apr 2023
- Adventist Youth – Church Leader Jan 2023 – Present
- Grade 11 With High Honors – GWA: 95 May 2022